

FPGA based Real Time Simulation of Power Converters

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Abstract: - The real-time simulation of high frequency power electronic converters on embedded platform is presented in this paper. The real-time simulation has the advantage of testing and studying real-time control algorithms. The recent developments in embedded technology and FPGA controllers have led to the real-time control and operation of many electronic apparatuses. Real-time control necessities real-time simulation in order to improve the control algorithm and to reduce the production time. The embedded systems have the potential of faster execution and thereby achieving real-time control as well as simulation. Power electronic converter are nowadays switched at a frequency more than 20KHz and upto 1MHz and so a faster simulation platform is needed. Embedded simulation of dc-to-dc converters using FPGA based Processor is developed in this work. The various input parameters like duty cycle, supply voltage, circuit components can be varied and the simulated output parameter is seen as an analog signal using a DAC interface.

Index terms: real-time simulation, power converters, FPGA Spartan Processor, DC-DC converters, embedded systems

I. INTRODUCTION

The simulation of the power electronic circuits is essential, as it is very expensive to study the dynamic nature of the circuits after building the same. The converters are mathematically modelled and simulated on conventional softwares [1]. Conventional simulation softwares require large amount of time for simulation. These simulations are implemented on a digital computer using a high-level language program to solve a system of differential equations in a time-stepping fashion. Real-time simulation can simulate the required parameter within the switching period itself. The current industrial approach to implement a digital controller is by using programmable devices such as FPGAs, microcontrollers, digital signal processors (DSPs) or microprocessors [2,3]. In all of these cases the control algorithm of the simulator is translated into the language of the target device. The advantage of this is that the controller runs at real-time. The real-time simulation needs an algorithm, which calculates and gives the solution of the differential equation of the system with an acceptable accuracy [4-7]. To implement this algorithm in an embedded FPGA controller, the parameters involved is to be scaled down for digital simulation [8,9]. The important challenge is in minimizing the execution time by using assembly level programming and by using fast ADC and DAC. The analog interface circuitry is used for input and

output interface to the real-time simulation system. To allow the simulation of the actual system to be carried out in real-time a high-speed embedded FPGA Spartan Processor is used [10-12].

II. ADVANTAGES OF REAL TIME SIMULATION

Power converters work at a switching frequency of above 20khz. So the simulation has to be done above 200khz. This limit ensures at least 10 samples within a switching period. Conventional simulation softwares require large amount of time for simulation. Real-time simulation can simulate the required parameter within the switching period itself [13]. The real-time simulation done on embedded FPGA controllers has the advantage of generating output in the form of analog signal using DAC interfaces. This helps in checking the output waveforms in a real-time basis. The real-time simulation on embedded system can be used to design and simulate controllers for closed loop control of system. After design of controllers, the same embedded system can be used for control applications also. This reduces the time between design and implementation of controllers for power electronic converters. The real-time simulation done on embedded system can be extended to hardware in loop simulation [14-16].

III. SYSTEM REQUIREMENTS

A. Hardware description

To simulate the converter system in real-time a separate processor system is required. The present system as shown in Figure 1 is based on real time simulation running on a FPGA Spartan Processor. The device on the target board features an internal oscillator which is enabled as the system clock source on reset.

The processor is a mixed-signal 32kb In-System Programmable (ISP) flash FPGA controller and features the following advantages.

- 12-bit, 100 kilo-Sample per second (KSPS) ADC.
- Two 12-bit DACs with 10us settling time. Two comparators configurable to generate interrupts on Reset.
- On-Chip JTAG emulation.

- 2304 Bytes data, RAM, 32k bytes FLASH, In-System Programmable in 512-byte sectors.

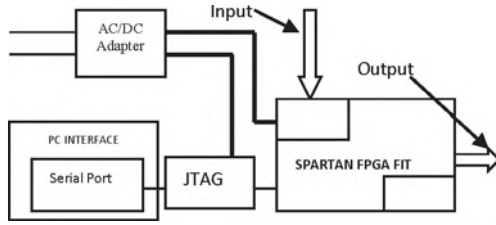


Fig. 1. Block Diagram of the Hardware Interface with FPGA Spartan 6 Processor

- Supply voltage – 2.7V to 3.6V
- Pipelined Instruction Architecture, Executes 70% of Instruction in 1 or 2 System Clocks.
- Reference – 2.4V, 15ppm/°C and Also External Reference Input.

B. Requirement for In-System Programming

The minimum system requirements are:

- Intel Pentium-class host PC running Microsoft Windows XP, Windows 7.
- One available COM port (1-4).
- 512 MB RAM and 40MB free Hard Disk space.

C. Software

Using the Integrated Development Environment (IDE) software, the entire FPGA Spartan Processor based target board is accessed. The IDE integrates a source-code editor, source-level debugger and in-system Flash programmer. This development kit includes the FPGA Spartan Processor macro assembler, linker and evaluation version 'C' compiler which can be used from with the IDE. The target board is interfaced with the personal computer with the serial adapter via the JTAG connectors of FPGA Spartan Processor.

IV. MODELING

A. Buck Converter Model

The Buck Converter as shown in Figure 2 with resistive (R) load is represented by mathematical modelling technique of Euler's Method.

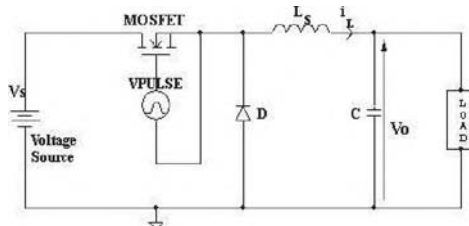


Fig. 2. Block Diagram of the Buck Converter

The equations are as follows:

Switch ON State:

$$i_L = i_{Lp} + \left((V_s - V_{sp}) * \frac{\Delta t}{L_s} \right) \quad (1)$$

$$V_{oi} = V_{sp} + \left(i_{Lp} * \frac{\Delta t}{C} \right) - \left(V_{sp} * \frac{\Delta t}{R * C} \right) \quad (2)$$

Switch OFF State:

$$i_L = i_{Lp} - \left(V_{sp} * \frac{\Delta t}{L_s} \right) \quad (3)$$

$$V_{oi} = V_{sp} + \left(i_{Lp} * \frac{\Delta t}{C} \right) - \left(V_{sp} * \frac{\Delta t}{R * C} \right) \quad (4)$$

The Buck Converter with resistive and inductive (RL) load is represented by mathematical modelling technique of Euler's Method. The equations are as follows:

Switch ON State:

$$i_s = (V_s - V_{sp}) * \frac{\Delta t}{L_s} + i_{sp} \quad (5)$$

$$V_{oi} = (i_{sp} - i_{sp}) * \frac{\Delta t}{C} + V_{sp} \quad (6)$$

$$i_o = (V_{sp} - R * i_{sp}) * \frac{\Delta t}{L_s} + i_{sp} \quad (7)$$

Switch OFF State;

$$i_s = (-V_{sp}) * \frac{\Delta t}{L_s} + i_{sp} \quad (8)$$

$$V_{oi} = (i_{sp} - i_{sp}) * \frac{\Delta t}{C} + V_{sp} \quad (9)$$

$$i_o = (V_{sp} - R * i_{sp}) * \frac{\Delta t}{L_s} + i_{sp} \quad (10)$$

B. Boost Converter Model

The Boost Converter as shown in Figure 3 with resistive (R) load is represented by mathematical modelling technique of Euler's Method.

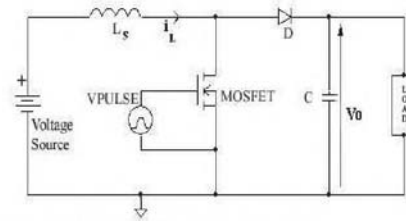


Fig. 3. Block Diagram of the Boost Converter

The equations are as follows:

Switch ON State:

$$i_L = \left(V_s * \frac{\Delta t}{L_s} \right) + i_{Lp} \quad (11)$$

$$V_{oi} = \left(V_{sp} * \frac{\Delta t}{R * C} \right) - V_{sp} \quad (12)$$

Switch OFF State:

$$i_L = \left((V_s - V_{sp}) * \frac{\Delta t}{L_s} \right) + i_{Lp} \quad (13)$$

$$V_{oi} = \left(i_{Lp} * \frac{\Delta t}{C} \right) - \left(V_{sp} * \frac{\Delta t}{R * C} \right) + V_{sp} \quad (14)$$

The Boost Converter with resistive and inductive (RL) load is represented by mathematical modelling technique of Euler's Method. The equations are as follows:

Switch ON State:

$$i_s = V_s * \frac{\Delta t}{L_s} + i_{sp} \quad (15)$$

$$V_{oi} = (-i_{op}) * \frac{\Delta t}{C} + V_{op} \quad (16)$$

$$i_o = (V_{op} - R * i_{op}) * \frac{\Delta t}{L_s} + i_{op} \quad (17)$$

Switch OFF State

$$i_s = (V_s - V_{op}) * \frac{\Delta t}{L_s} + i_{sp} \quad (18)$$

$$V_{oi} = (i_{op} - i_{sp}) * \frac{\Delta t}{C} + V_{op} \quad (19)$$

$$i_o = (V_{op} - R * i_{op}) * \frac{\Delta t}{L_s} + i_{op} \quad (20)$$

Where

i_{sp} is the Previous Value of Inductor Current.

i_s is the Present value of Inductor Current

V_{op} is the Previous Value of Output Voltage

V_{oi} is the Present Value of Output Voltage

i_{op} is the Previous Value of Output Current

i_o is the Present Value of Output Current

L_s is the Circuit inductor

L is the Load Inductance

C is the Circuit Capacitance

R is the Load Resistor

Δt is the Sampling Interval

V. SIMULATION AND EXPERIMENTAL RESULTS

The buck converter system equations were simulated first using 'C' language. The program was then transformed into Assembly Language of FPGA Spartan Processor. The intel processor compatible *.hex" file was downloaded into the 32KB Flash Memory of the processor using the Flash Utility. The program for the initialization of the Special Function Registers (SFR) were included in the "include" file of the processor. The processor was initialized to operate with the internal clock of 16MHz frequency. The program disables the watchdog timer first to keep all internal registers and memories maintain their original data. The ports were configured to the states. Then the internal voltage reference was enabled. The digital to analog converter ports was initialised with data. In order to make the simulation faster, so that it will be on the real-time, steps were taken to minimize the number of instructions in the calculation of output voltage and inductor current at each iteration and signal output loop.

A. Experimental Setup

The experimental setup consists of the FPGA Kit connected with the PC as shown in the Figure 4. The output from the FPGA processor is connected to the digital storage oscilloscope. The simulation software MICROSIM is installed on a separate PC with the same simulation parameters.



Fig. 4. Hardware setup of the project

B. Algorithm

- Define converter parameters and its values.
- Decide the sampling interval, Δt based on the switching frequency. This is important to have real-time simulation.
- Get the values of the variables of the converter viz., duty cycle, the inductance L and the capacitance C .
- Find the ON time and OFF time periods for the given duty cycle and switching frequency.
- Find the number of iterations required for the ON time and the OFF time using sampling interval.
- Define and calculate the constants related with the output voltage and inductor current equations.
- Initialize the Processor – its ports, reference voltage and timers.
- Initialize the Digital to analog converter ports.
- Calculate the output voltage and inductor current corresponding to ON period and output it on the analog interface. Wait for the sampling interval and repeat this for the number of times, decided by ON period.
- Repeat the above step for the OFF period and the number of times decided by OFF period.
- Continuously repeat the above steps until halted by the user.

The simulation is done with three different frequencies of 1kHz to 3 kHz with source voltage of 10V, inductance of 15mH and load resistance of 10Ω. The capacitance is varied between 10μF to 1000μF. The simulation results of the buck converter simulated through PC-based software simulation and FPGA-based hardware simulation is presented in this paper in the Figure 5 to Figure 20. The results show the latency in both the systems and their effects in the real hardware simulation.

C. Results and Conclusion

The simulation of the Buck converter was done on the FPGA Spartan Processor based system and the same was simulated with PSPICE-based simulation software. The output voltage for the changes in the circuit parameters was

simulated. The time required for processing of power converters at real-time was analysed. The graph of the analog output of the Real-Time Simulation and the PSPICE-based simulated results are given in Figure 5 to Figure 20. It is inferred from the results that the increase in the capacitance value and the switching frequency reduces the ripples in the output voltage. The Real-Time simulations results were compared with the PSPICE-based simulation. Further investigations are being done for the increasing the switching frequency and reducing the execution time with the optimization of the algorithm.

REFERENCES

- [1] Muhammad H. rashid "Power Electronics: Circuits, devices and application", 2nd Ed., Prentice Hall, July 2002, pp. 303 – 351.
- [2] Myke Predko "Programming and customizing the 8051 micro controller", Tata McGraw-Hill Ed., 1999, pp. 37 – 199.
- [3] N. Senthilkumar, "Real-Time Simulation" presented at PCIC 2003 – National Level Conference, Palakkad.
- [4] Christian Dufour, Jean Belanger, Simon Abourida, "Accurate Simulation of a six-pulse Inverter with Real-Time Event Compensation", presented at Electrimacs 2002 – International Conference, August 18-21, 2002
- [5] Jean Belanger, Nicolas lechevin, Guillaume Murere, "Real-Time Simulation of Averaged Models of Power Converters (Part I,II,III)" presented at Electrimacs 2002 – International Conference, October 26, 2002.
- [6] Venkata R. Dinavahi, M. Reza Iravani, and Richard Bonert, "Real-Time Digital Simulation of Power Electronics Apparatus Interfaced with Digital Controllers", Presented in IEEE Transaction on Power Deliver, Vol. 16, No. 4, October 2001 – pp 775 – 781.
- [7] H. J. Slater, D. J. Atkinson, A. G. jack, "Real-Time Emulation for Power Equipment Development. Part 2: Real-Time Simulation", Presented at IEEE Proc – Electrical Power Appl., Vol. 145, No. 3, May 1998 – pp 153 – 158.
- [8] H. J. Slater, D. J. Atkinson, A. G. jack, "Real-Time Emulation for Power Equipment Development. Part 1: Real-Time Simulation", Presented at IEEE Proc – Electrical Power Appl., Vol. 145, No. 2, March 1998 – pp 92 – 97.
- [9] Alfred Fiedler and Horst Grotstollen, "Simulation of Power Electronic Circuits with Principles used in Wave Digital Filters", Presented at IEEE Transaction on Industry Appl., Vol. 33, No. 1 Jan – Feb 1996.
- [10] Jiménez Ó., Lucía Ó., Urriza I., Barragan L.A., Navarro D., Dinavahi V. "Implementation of an FPGA-Based Online Hardware-in-the-Loop Emulator Using High-Level Synthesis Tools for Resonant Power Converters Applied to Induction Heating Appliances." IEEE Trans. Ind. Electron. 2015, 62, 2206–2214.
- [11] Yan Q., Tasiu I., Chen H., Zhang Y., Wu S., Liu Z., "Design and Hardware-in-the-Loop Implementation of Fuzzy-Based Proportional-Integral Control for the Traction Line-Side Converter of a High-Speed Train." Energies 2019, 12, 4094.
- [12] Faruque O.M., Strasser T., Lauss G., Jalili-Marandi V., Forsyth P., Dufour C., Dinavahi V., Monti A., Kotsampopoulos P., Strunz K., "Real-Time Simulation Technologies for Power Systems Design, Testing, and Analysis." IEEE Power Energy Technol. Syst. J. 2015, 2, 63–73.
- [13] Li W., Belanger J., "An Equivalent Circuit Method for Modelling and Simulation of Modular Multilevel Converters in Real-Time HIL Test Bench." IEEE Trans. Power Deliv. 2016, 31, 2401–2409.
- [14] Liu, R. Ma, H. Bai, Z. Li, F. Gechter, F. Gao, "Hybrid modeling of power electronic system for hardware-in-the-loop application," Electr.Power Syst. Res., vol 163, pp. 502-512, Oct. 2018.
- [15] M. Milton and A. Benigni, "Latency Insertion Method Based Real-Time Simulation of Power Electronic Systems," IEEE Trans. Power Electron., vol. 33, no. 8, pp. 7166-7177, Aug. 2018.
- [16] Wentao Wang, Zhuzuan Shen, and Venkata Dinavahi, "Physics-Based Device-Level Power Electronic Circuit Hardware Emulation on FPGA", IEEE Transactions on Industrial Informatic, Vol. 10, NO. 4, November, 2014.

A. REAL TIME SIMULATION RESULTS

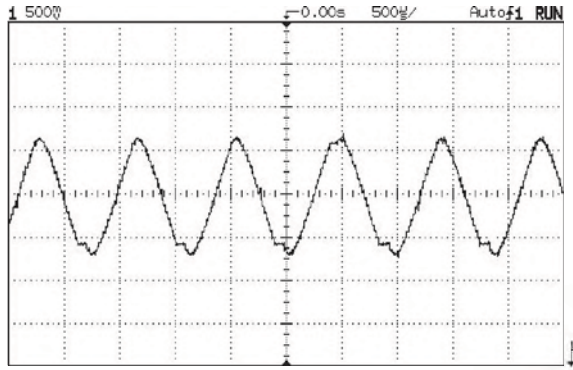


Fig. 5. Buck Converter with Freq=1Khz, Vs=10v, L=15mH, C=10uF, R=10Ω

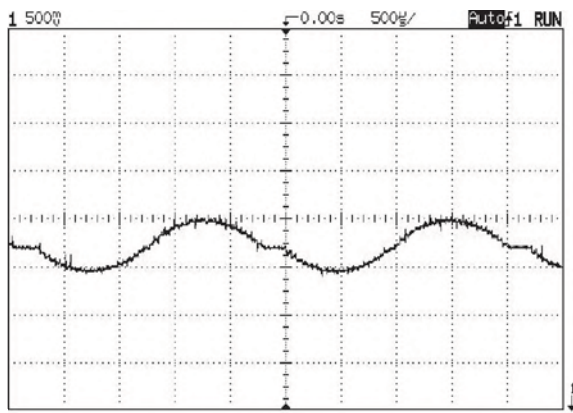


Fig. 6. Buck Converter with Freq=1Khz, Vs=10v, L=15mH, C=100uF, R=10Ω

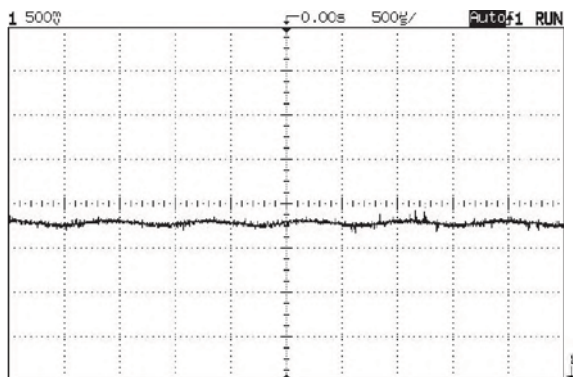


Fig. 7. Buck Converter with Freq=1Khz, Vs=10v, L=15mH, C=1000uF, R=10Ω

B. MICROSIM SIMULATION

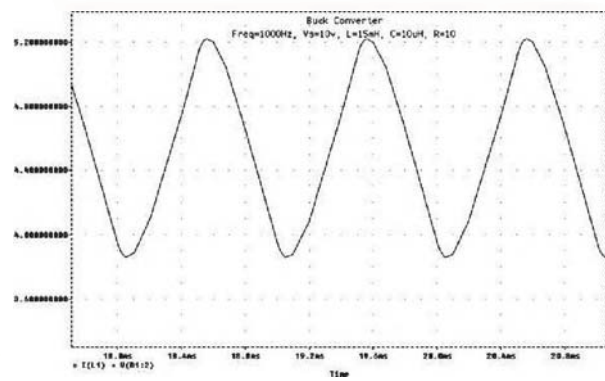


Fig. 8. Buck Converter with Freq=1Khz, Vs=10v, L=15mH, C=10uF, R=10Ω

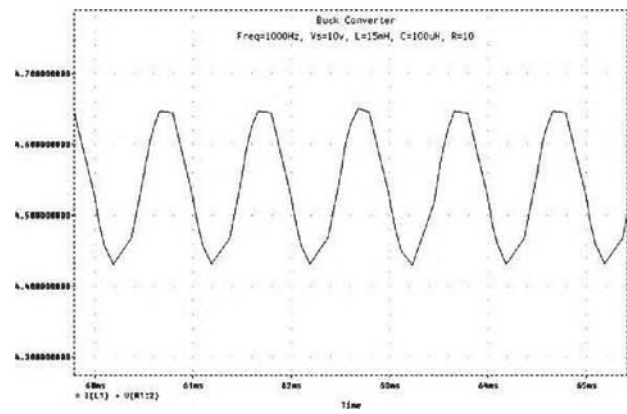


Fig. 9. Buck Converter with Freq=1Khz, Vs=10v, L=15mH, C=100uF, R=10Ω

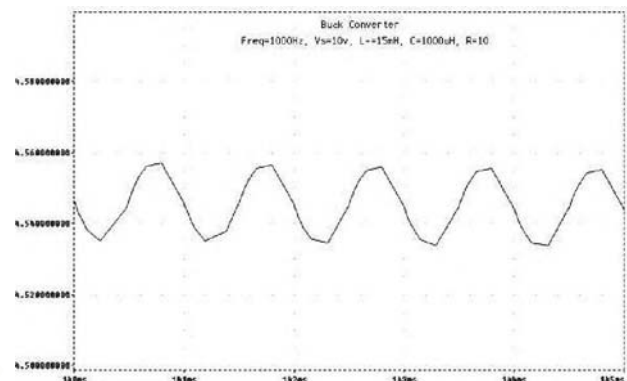


Fig. 10. Buck Converter with Freq=1Khz, Vs=10v, L=15mH, C=1000uF, R=10Ω

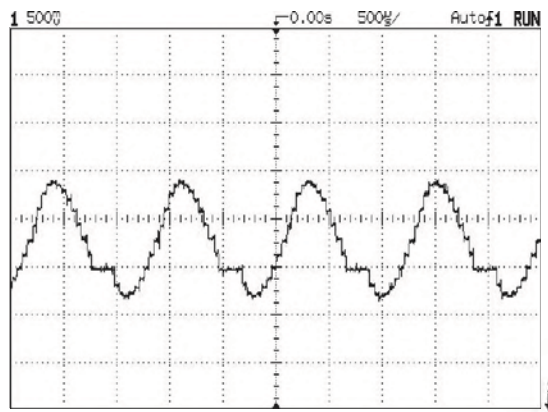


Fig. 11. Buck Converter with Freq=2Khz, $V_s=10v$, $L=15mH$, $C=10\mu F$, $R=10\Omega$

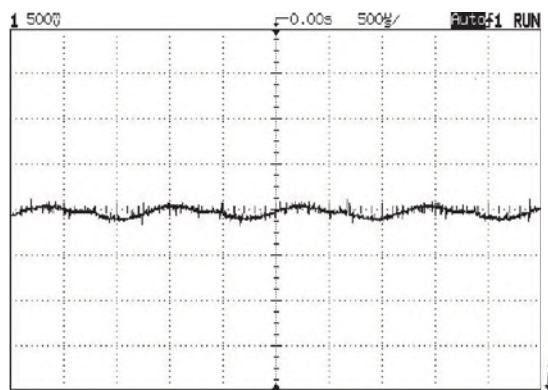


Fig. 12. Buck Converter with Freq=2Khz, $V_s=10v$, $L=15mH$, $C=100\mu F$, $R=10\Omega$

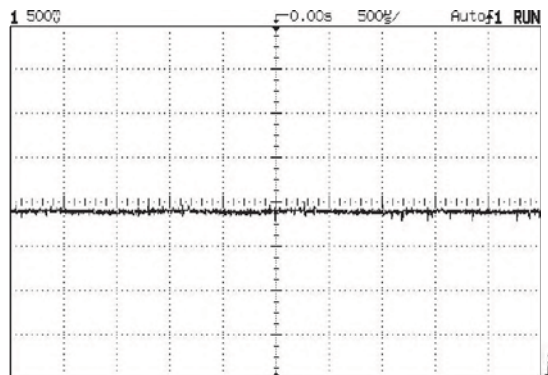


Fig. 13. Buck Converter with Freq=2Khz, $V_s=10v$, $L=15mH$, $C=1000\mu F$, $R=10\Omega$

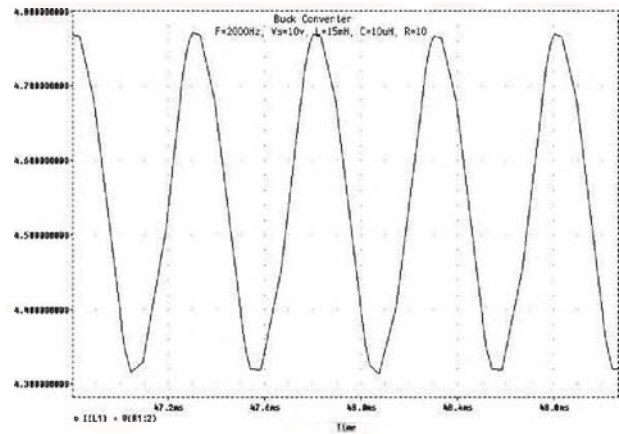


Fig. 14. Buck Converter with Freq=2Khz, $V_s=10v$, $L=15mH$, $C=10\mu F$, $R=10\Omega$

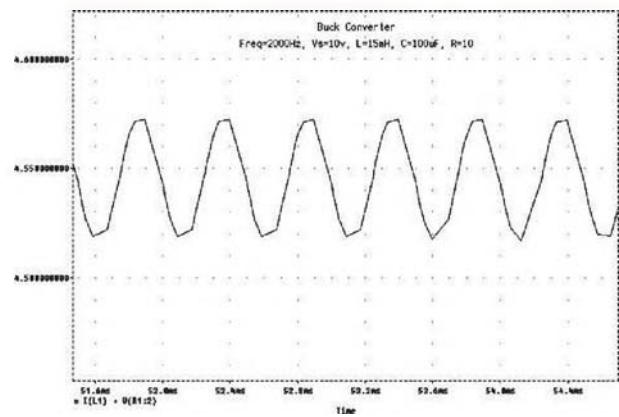


Fig. 15. Buck Converter with Freq=2Khz, $V_s=10v$, $L=15mH$, $C=100\mu F$, $R=10\Omega$

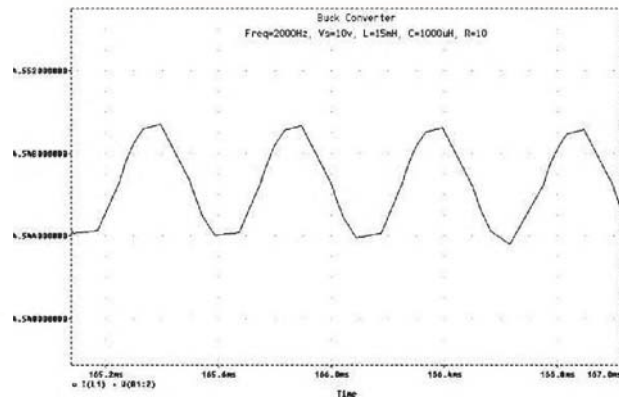


Fig. 16. Buck Converter with Freq=2Khz, $V_s=10v$, $L=15mH$, $C=1000\mu F$, $R=10\Omega$

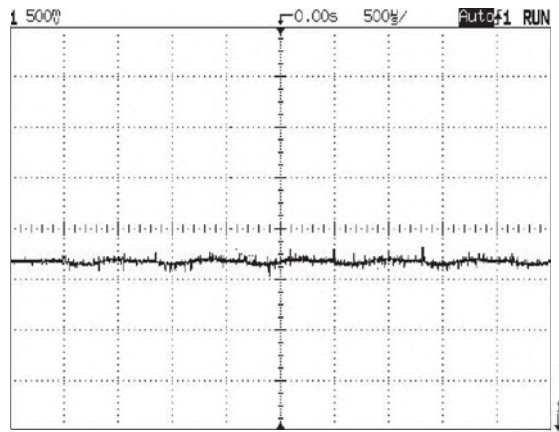


Fig. 17. Fig. 10. Buck Converter with Freq=3Khz, $V_s=10v$, $L=15mH$, $C=100\mu F$, $R=10\Omega$

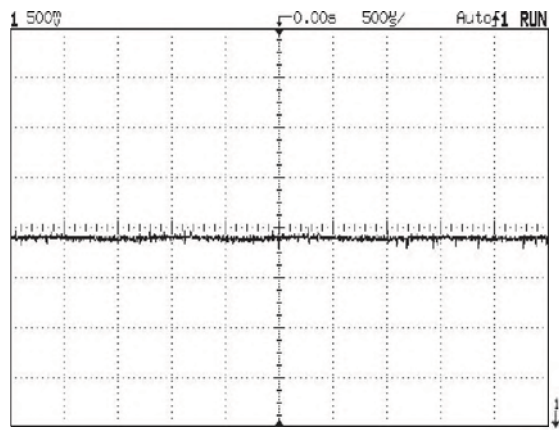


Fig. 18. Fig. 11. Buck Converter with Freq=3Khz, $V_s=10v$, $L=15mH$, $C=1000\mu F$, $R=10\Omega$

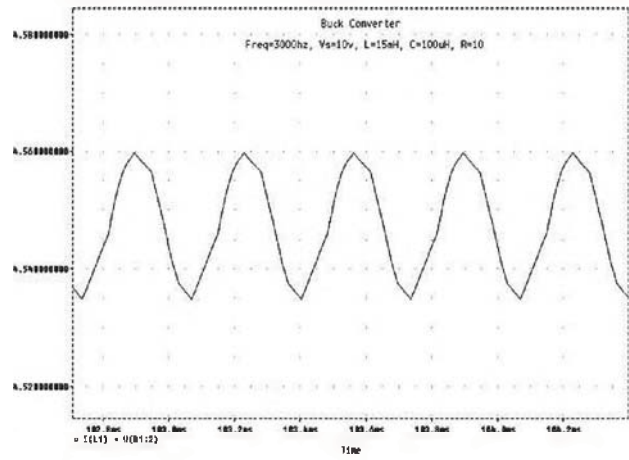


Fig. 19. Fig. 10.a. Buck Converter with Freq=3Khz, $V_s=10v$, $L=15mH$, $C=100\mu F$, $R=10\Omega$

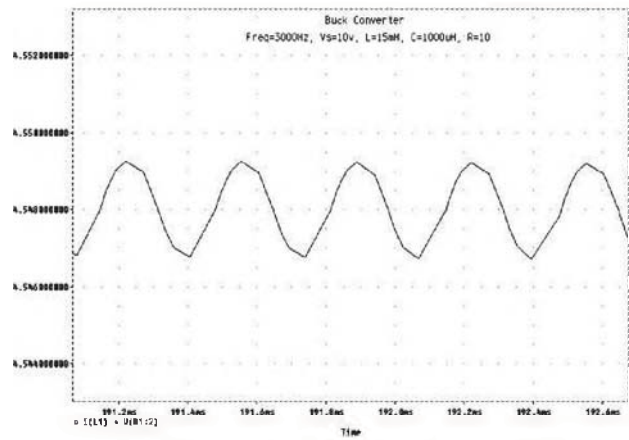


Fig. 20. Fig. 11. Buck Converter with Freq=3Khz, $V_s=10v$, $L=15mH$, $C=1000\mu F$, $R=10\Omega$